

Knowing When to Revise: Confidence-Gated Self-Correction in LLMs

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github.com/Fakharalamkhan/confidence-gated-self-correction



1 Hypothesis

Problem

- Intrinsic self-critique often turns right answers into wrong ones [1, 2].

Idea

Revise only if the confidence in A_0 is low:

$$\text{revise} \Leftrightarrow S(A_0) < \tau$$

RQs and hypotheses

RQ1 Does a gate beat never- and always-revise?

H1 Yes, it beats both.

RQ2 Which signal best finds wrong answers?

H2 Self-consistency.

RQ3 How large can the gain of a gate be?

H3 It is capped by the fix rate f :

$$\text{Acc} = a + (1-a) f t - a b u$$

a initial accuracy

f fix rate

t share of wrong answers revised

b break rate

u share of right answers revised

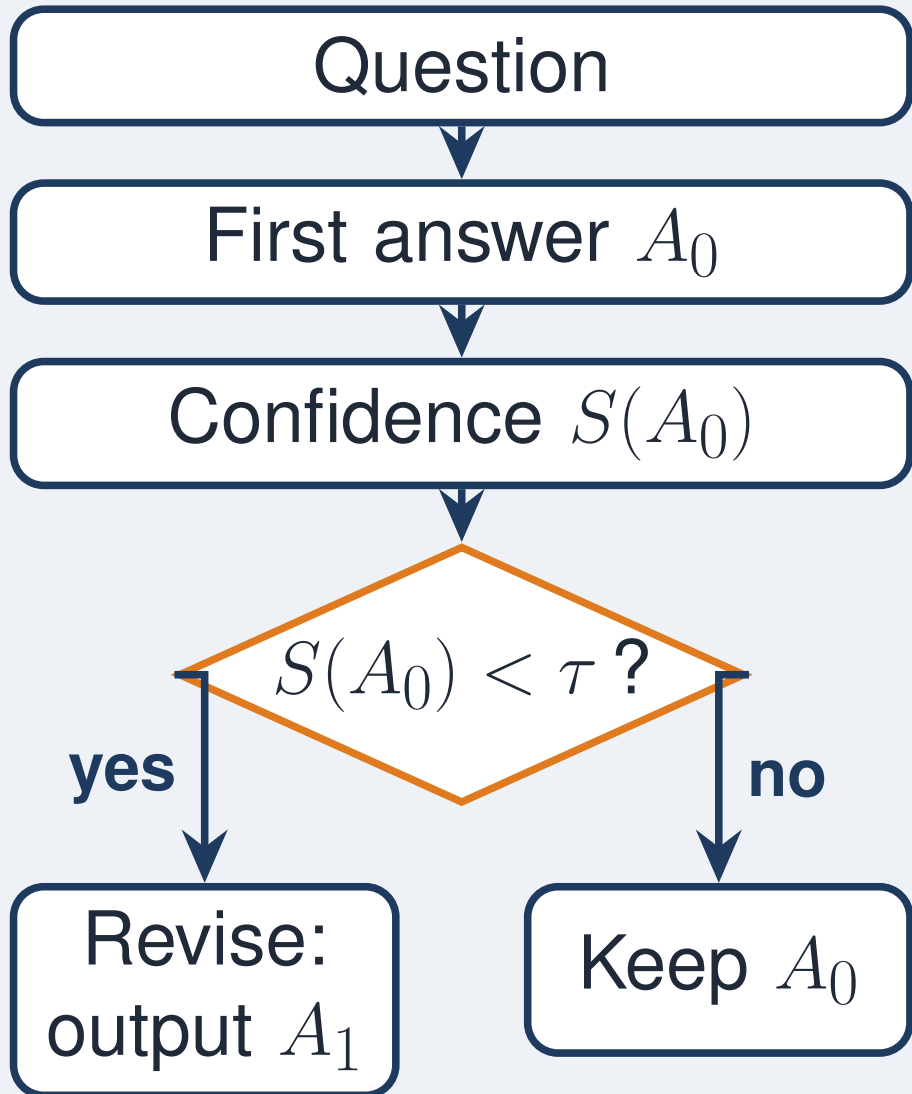
Ceiling on the gain: $(1-a) f$

Prior work

- IoE: keep or revise inside one prompt [3]
- Confidence vs. critique ability [4]
- P(True) self-verification [5]
- Self-consistency by sampling [6]
- New here: explicit threshold gates, tuned on held-out data

2 Methodology

Pipeline



Setup and why

- Qwen2.5-7B-Instruct [9]: open, fits 2x NVIDIA T4 with vLLM [10]
- GSM8K [7] (math), HotpotQA [8] (multi-hop QA): answers checked automatically
- 100 dev / 400 test each: τ never tuned on test

Signals

One from each family in prior work:

- Verbalised confidence S_{verb}
- Self-consistency S_{sc} , $N=10$ samples [6]
- P(True) S_{verif} [5]

Baselines, metrics

- Never, always, majority vote, IoE [3]; oracle as upper bound
- Majority vote as revision step: a step that does fix errors
- $W \rightarrow R$ / $R \rightarrow W$ instead of net accuracy
- AUROC, token cost; paired exact McNemar test

Code and data

github.com/Fakharalamkhan/confidence-gated-self-correction

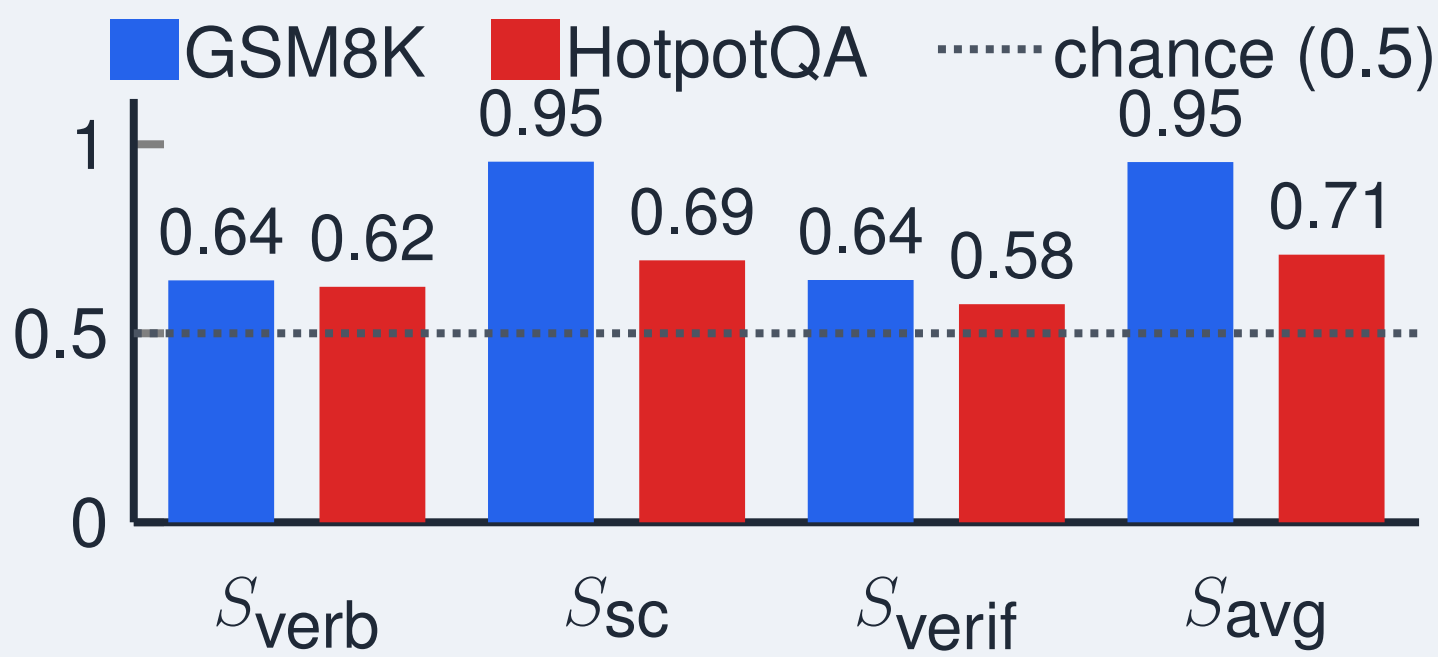
3 Results

(a) Always revise: transitions

Dataset	$W \rightarrow R$	$R \rightarrow W$	f	b	κ
GSM8K	0	1	0.000	0.003	∞
HotpotQA	6	21	0.036	0.089	3.50

400 test questions per dataset; κ = breaks per fix.

(b) AUROC per signal



(c) Test accuracy (%)

Self-critique Method	GSM8K		HotpotQA	
	Acc	p	Acc	p
Never revise	92.5	—	58.8	—
Always revise	92.3	1.00	55.0	0.006
Majority vote	95.5	0.004	60.0	0.383
IoE	93.0	0.500	57.3	0.070
Best gate (dev) [†]	92.5	1.00	58.8	1.00
Oracle	92.5	1.00	60.3	0.031

p vs. never revise (McNemar). [†] All gates tie.

(d) Majority vote as the revision step

Dataset	Fixes	Breaks	κ	S_{sc} gate
GSM8K	14 ($f=0.47$)	3	0.21	92.5 %
HotpotQA	18	14	0.78	59.5 %

59.56 %

59.50 %

predicted by the model observed on HotpotQA

(e) Verdict on the hypotheses

- ✗ **H1**: no dev-tuned gate beats never-revise
- ✓ **H2**: AUROC S_{sc} 0.95 / 0.69; S_{verb} , $S_{\text{verif}} \leq 0.64$
- ✓ **H3**: ceiling 0 / 1.5 points, never exceeded

(f) Unexpected outcomes

- GSM8K: self-critique changed only 2 of 500
- GSM8K: P(True) near 0 even when correct

(g) Limitations

- One 7B model; 400 test questions each
- One revision prompt; dev split of 100

Take-home: the bottleneck is the revision step, not the gate.